

IE 469 INDUSTRIAL APPLICATIONS OF OPERATIONS RESEARCH

FALL 2025 TERM PROJECT

Project Due: December 11 at 23:59 (Sharp!). **All members should submit their groups work individually. Failure to submit this project will result in an FZ grade.**

Deliverables: You should work with your Case Study groups. **(1)** Each group must submit an Excel file containing all components of the project **(2)** You should also write a report that should have the details as to how you handle the project (your mathematical models, algorithms, assumptions... etc.)

Project Demo: Each group will do a 10-minute demo to the course TA during the lecture hours. (Dates and times will be announced later)

Grading: There is NO specific requirement as to a specific method (like a mathematical model or a heuristic algorithm) has to be used to do this project. You are free to choose any methodology you want as long as you address the problem properly and provide a satisfactory and reasonable solution. **The grading will be based on your effort, and the quality, usability and flexibility of the final product among other groups.**

Academic Dishonesty: This project must be done solely by your group and by your group only. If we suspect any dishonesty, your project grade will automatically be zero and there will be disciplinary action. This includes any code you may borrow from the Internet (this obviously includes Generative AI tools) without proper citation! (If you cite your source, you are free to use any code from the Internet.)

Disclaimer: This project story is purely fictitious. Any relation to a real life event or incident is purely coincidental! All images and logos belong to the corresponding owner.

“THE DINNER IS LATE, IT AIN’T NO GREAT!”



New York City has one of the most competitive and dynamic food-delivery markets in the world. The combination of high population density, diverse restaurant offerings, and consumers’ growing preference for on-demand convenience has made delivery apps a critical part of the city’s food economy. Major players — Uber Eats, DoorDash, Grubhub — compete to secure restaurant partnerships and customer loyalty.

These companies operate a platform-based, on-demand delivery model connecting three primary stakeholders: customers, restaurants, and delivery partners (drivers). Customers use the app to place an order from a nearby restaurant within a given distance from the customers’ location (called service area coverage). When a customer orders from the platform, it is assigned to one of the available drivers based on proximity, transport mode (bike, scooter, car), and predicted travel time. These drivers are independent contractors paid per delivery and usually do not typically have fixed working hours. In the meantime, the restaurant receiving the order starts the preparation. Once the order is ready, the driver picks it up from the restaurant and delivers it to the customer.

There is a heavy usage of these apps in New York City as 48% of people of ages 25-34 and 63% of the people of 18-24 use these online delivery apps. However, only 13% of people older than 35 use these apps. Moreover, overall average across third-party apps is approximately 4.6 orders/month per user.

Among these companies, Uber Eats has some issues related to on time delivery performance in New York’s dense urban environment. Traffic congestion with speeds averaging to as low as 9 mph and finding an available delivery driver in proximity could cause excessive delivery times and inconsistent reliability. There are several reports highlight systemic issues that affect Uber Eats deliveries:

- In a Business Insider article, Uber Eats announced that as of May 28, 2024 the company would discontinue the option for couriers to deliver orders *on foot* in New York City, citing inefficiencies and new pay-law constraints. This change

raised concerns that walking deliveries — which in dense urban blocks might be the fastest mode — were being eliminated, potentially exacerbating delivery times. [Business Insider](#)

- On Reddit threads and review forums, numerous users recount long delays with Uber Eats orders: one user reported waiting three hours for a delivery and being offered only a minimal refund. [The Sun](#)
- Expert commentary identifies primary causes for delays including driver acceptance rates, batching of orders, restaurant preparation bottlenecks, heavy traffic and urban access issues. [Marketing Scoop](#)

Officials at Uber Eats wish to address the above-mentioned issues by reviewing the sources of delay in their delivery process, and found three major sources:

- I. Delay due to no available drivers at the time of order.
- II. Delay due to unready order at the restaurant.
- III. Delay due to traffic congestion.

As they have no direct control over the individual restaurants or the traffic conditions, they focus on the delay due to driver availability. In order to maintain driver availability, they wish to **dynamically adjust driver allocation and service area coverage** per each Neighborhood Tabulation Area (NTA) of Manhattan to improve service quality. In particular, they would like to establish a systematic approach to **change the service area coverage** for each NTA based on the number of drivers available and the expected traffic conditions so that once an order is placed, there will be an available driver that can immediately be assigned to it. For this purpose, you are given the demographic information and geographic boundaries of each NTA and a restaurant dataset of Manhattan.

- 1) (30 pts.)** Determine the set of restaurants in the service area coverage of each NTA given the radius of coverage.

Input: NTA and its radius of coverage (in miles)

Output: All the distances (in miles) of all restaurants to the NTA within its radius of coverage (Calculating rectilinear or “Manhattan” distances is preferred.)

- The radius should emanate from the geographic center point of the NTA.
- Based on the radius given, the coverage could entirely remain within the NTA or overlap with other NTAs.

- 2) (40 pts.)** Determine the driver allocation and service area coverage per NTA that would satisfy driver availability for each new order.

Input: Total number of drivers available in Manhattan.

Output: Driver allocation and service area coverage per NTA.

Assumptions:

- Normally, drivers are supposed to be “re-allocated” as the number of available drivers change throughout the day. However, the allocation should remain at the level of determining just the amount, not making individual driver assignments to NTAs. So, you will determine how many drivers out of the total number available should serve each NTA.
- A driver can only handle one order at a time.
- Regardless of the mode of travel, time of day, or day of week; drivers travel at 9 mph.
- Orders from the customers are distributed uniformly among the lunch and dinner hours of a day, any day of a month. Lunch hours are 11:30-13:30, dinner hours are 17:30-21:00. No orders are received outside these time slots.
- Customers pick a restaurant within the service area coverage of their NTA with an equal probability.
- Customers use one of the three apps (Uber Eats, Grubhub and DoorDash) with an equal probability.
- Customer locations are uniformly distributed within the NTAs.

- 3) (30 pts.)** Develop a decision support system utilizing UserForms (not just message boxes) that will take the total number of drivers available and run the above methodology. It should be able to provide the output of the above methodology displayed preferably on a map screen.

Feel free to make any other reasonable assumptions if you need. Do not forget to have all these additional assumptions in your report.